Simulation Study on Improving Spectrally Resolved Solar-Induced Fluorescence Satellite Retrievals Under Uncertain Atmospheric States

Antti Kukkurainen, Antti Lipponen, Neus Sabater, Antti Arola, and Ville Kolehmainen

***Abstract*— Over recent decades, advancements in satellite remote sensing for solar-induced fluorescence (SIF) have accelerated, driven by the link between SIF and vegetation photosynthesis. Many algorithms focus on SIF at specific spectral regions, such as Solar Fraunhofer lines or oxygen absorption bands at the top-of-atmosphere (TOA) and top-of-canopy (TOC) levels, respectively. However, few algorithms capture SIF across the entire red-to-near-infrared range. The primary challenge in estimating total emitted SIF at the TOA level is accurately disentangling atmospheric disturbances. This article introduces the approximation error (AE) method to model and compensate for the modeling uncertainties associated with the atmospheric scattering and absorption effects. Coupled with the Bayesian retrieval algorithm called SIFFI, developed to estimate spectrally resolved SIF, the AE technique, tested on simulated data, can be generalized and integrated with other SIF retrieval approaches. This integration significantly improves SIF estimates, particularly when atmospheric conditions are not well-characterized.**

***Index Terms*— Approximation error (AE), Bayesian inversion, radiative transfer, retrieval algorithm, satellite observations, solar-induced chlorophyll fluorescence (SIF).**

NOMENCLATURE

SIF	Solar-induced fluorescence.
TOA	Top-of-atmosphere.
TOC	Top-of-canopy.
AOD	Aerosol optical depth.
MAP	Maximum a posteriori.
AE	Approximation error.
LUT	Lookup table.
TCWV	Total column water vapor.
SSA	Single-scattering albedo.

Received 10 January 2025; revised 6 May 2025 and 3 July 2025; accepted 13 July 2025. Date of publication 18 July 2025; date of current version 28 July 2025. This work was supported in part by the University of Eastern Finland (UEF) Doctoral Programme in Science, Technology and Computing (SCITECO) under Grant 203/03.01.02/2020; in part by the Research Council of Finland (ArcticSIF) under Grant 356937; in part by the Research Council of Finland Flagship of Atmosphere and Climate Competence Center (ACCC) under Grant 357904; in part by the Research Council of Finland Flagship of Advanced Mathematics for Sensing, Imaging and Modeling (FAME) under Grant 358944; and in part by the Research Council of Finland Centre of Excellence in Inverse Modelling and Imaging under Grant 353084. *(Corresponding author: Antti Kukkurainen.)*

Antti Kukkurainen is with the Finnish Meteorological Institute, Atmospheric Research Centre of Eastern Finland, 70211 Kuopio, Finland, and also with the Department of Technical Physics, University of Eastern Finland, 70211 Kuopio, Finland (e-mail: antti.kukkurainen@fmi.fi).

Antti Lipponen, Neus Sabater, and Antti Arola are with the Finnish Meteorological Institute, Atmospheric Research Centre of Eastern Finland, 70211 Kuopio, Finland (e-mail: antti.lipponen@fmi.fi; neus.sabater@fmi.fi; antti.arola@fmi.fi).

Ville Kolehmainen is with the Department of Technical Physics, University of Eastern Finland, 70211 Kuopio, Finland (e-mail: ville.kolehmainen@uef.fi).

This article has supplementary downloadable material available at https://doi.org/10.1109/TGRS.2025.3590496, provided by the authors.

Digital Object Identifier 10.1109/TGRS.2025.3590496

I. INTRODUCTION

SIF is a low-intensity radiation signal emitted by plants' chlorophyll molecules in the red and far-red spectral ranges (650–800 nm) released alongside photochemical processes as part of the plant's response to absorbed solar energy [1]. SIF monitoring has lately gained attention due to its suggested use as a proxy for photosynthetic activity and gross primary production (GPP) [2].

Over the past few decades, significant advancements have been made in the development of SIF retrieval techniques [3]. While early SIF retrievals revolved around ground- and airborne-based methods, the latest developments in satellite remote sensing have enabled global monitoring of SIF [4], [5]. Techniques such as the Fraunhofer line discrimination (FLD) method applied mainly to the oxygen O_2-A band [6], [7] have been central to the success of SIF retrievals in the past for ground-level applications. Today, SIF data products are an integral part of various satellite missions, exploiting the Solar Fraunhofer bands in the far-red region. Examples of these are the Orbiting Carbon Observatory [OCO-2(3)] [8], [9], the Greenhouse gases Observing SATellite (GOSAT) [10], and the Sentinel-5 Precursor (S-5P) [11], [12] datasets.

At the TOA level, the satellite-observed radiance signal accounts for the contributions from the atmosphere, surface, and small fluorescence emission. To minimize the impact of atmospheric disturbance, most SIF retrieval algorithms at the TOA level quantify fluorescence at selected narrow solar absorption lines [13], [14]. These lines together with bands which miss the strong atmospheric absorption characteristics can be exploited as fitting windows to reconstruct the full SIF spectrum as shown recently in [15] using TROPOMI observations. However, across the red-to-near-infrared region, the primary atmosphere-related error sources overlapping the fluorescence emission include atmospheric absorption by water vapor, oxygen, and the presence of absorbing and scattering aerosols [16]. These disturbances complicate any total SIF retrieval attempt, as an inaccurate knowledge of the atmospheric characterization impacts the SIF estimates.

With the goal of achieving total SIF retrievals—especially in light of upcoming hyperspectral missions like Fluorescence Explorer (FLEX) [17]—we recently introduced a novel method called SIFFI [18] for retrieving SIF across the red-to-near-infrared spectrum at both TOC and TOA levels, the

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

latter depending on the availability of auxiliary atmospheric information. However, fluorescence estimates from any SIF retrieval technique applied to atmospherically corrected data will highly depend on the accuracy achieved at the previous atmospheric correction step. To tackle such a bottleneck, we explore the possibility of coupling the AE technique [19], [20], which is based on the marginalization of the poorly known auxiliary parameters in the forward modeling, to an SIF retrieval technique, in this case, the SIFFI method.

When uncertainties are associated with the atmospheric state, the AE method is able to mitigate the resulting modeling error in the retrieval process, thus improving the accuracy of SIF retrievals from TOA observations. The possibility of coupling the AE method to different retrieval techniques expands the opportunities for the expected upcoming algorithm developments as a response to the increasing hyperspectral missions planned with the potential to provide SIF estimates at wider spectral regions, e.g., CO2M [21].

In this article, we provide, as a proof-of-concept, a simulation-based assessment of the coupled SIFFI AE method at the TOA level when the atmospheric state is not accurately known.

II. Materials and Methods

A. SIFFI Retrieval Algorithm

We give a brief summary of the SIFFI retrieval presented in [18]. Assuming a Lambertian surface and an isotropic fluorescence emission, the forward model for satellite-measured spectral radiance at TOA can be written as

$$L_{\text{sim}}(\rho_{\text{surf}}, F, \eta, \zeta) = L_0(\eta, \zeta) + \frac{\rho_{\text{surf}} \cdot E_{\text{toc}}(\eta, \zeta) \cdot T^{\uparrow}(\eta, \zeta) + \pi F \cdot T^{\uparrow}(\eta, \zeta)}{\pi(1 - \rho_{\text{surf}} \cdot S(\eta, \zeta))} \tag{1}$$

where L_{sim} is the simulated radiance at TOA, ρ_{surf} is the surface reflectance, F is the SIF, L_0 is the atmospheric path radiance, E_{toc} is the total solar irradiance at TOC, $T^{\uparrow}$ is the total upward transmittance, and S is the spherical albedo [22], [23]. The parameter vector η in (1) denotes the observation geometry-related parameters such as the solar and sensor view zenith angles, and the parameter vector ζ denotes the atmosphere-related parameters such as the AOD and the total water vapor content in the atmospheric column.

In SIFFI, we follow the Bayesian framework for inverse problems and model all the unknown model variables as random variables where the uncertainty about the values of the variables is coded in the models of their probability distributions [19]. The observation noise is usually modeled as an additive zero-mean Gaussian noise with a noise covariance Γ_e, leading to observation model

$$H_{\text{obs}} = L_{\text{sim}} + e \tag{2}$$

where $H_{\text{obs}} \in \mathbb{R}^n$ is the vector of observed radiances (measured at n wavelengths) and $e \sim \mathcal{N}(0, \Gamma_e)$ denotes the measurement noise. Conventionally, the observation geometry η and possibly the atmospheric state ζ are treated as known, fixed parameters in SIF estimation. Let θ denote a vector

$$\theta = \begin{bmatrix} \rho_{\text{surf}} \\ F \end{bmatrix} \in \mathbb{R}^{2n \times 1} \tag{3}$$

containing unknown surface reflectances and SIF for the n wavelengths. Using Bayes' theorem, the posterior probability density model for the unknown parameters can be written as

$$p(\theta|H_{\text{obs}}) = \frac{p(H_{\text{obs}}|\theta)p(\theta)}{p(H_{\text{obs}})} \tag{4}$$

where $p(H_{\text{obs}}|\theta)$ is the likelihood density which models the probability density of observing the data $p(H_{\text{obs}})$ given the data θ, $p(\theta)$ is the prior probability density which models prior knowledge about the unknown θ, and $p(H_{\text{obs}})$ is the probability density for the observed H_{obs} which can be treated as a normalization constant in this case. Thus, the posterior density model becomes

$$p(\theta|H_{\text{obs}}) \propto p(H_{\text{obs}}|\theta)p(\theta) \tag{5}$$

where the likelihood model $p(H_{\text{obs}}|\theta)$ under the noise model (2) can be written as

$$p(H_{\text{obs}}|\theta) = p_{e|\theta}(H_{\text{obs}} - L_{\text{sim}}|\theta) \tag{6}$$

where $p_{e|\theta}(H_{\text{obs}} - L_{\text{sim}}|\theta)$ denotes the likelihood model which has been marginalized with respect to the unknown and uninteresting measurement error e. Under the usual assumption that the measurement errors e are uncorrelated with the unknown parameters ρ_{surf} and F, and using a Gaussian prior model $p(\theta) \sim \mathcal{N}(\mu_\theta, \Gamma_\theta)$, the MAP estimate amounts to solving the optimization problem

$$\begin{aligned} \hat{\theta}_{\text{MAP}} &= \arg\max_{\theta} \; p(\theta|H_{\text{obs}}) \\ &= \arg\min_{\theta} \; \|K_e(H_{\text{obs}} - L_{\text{sim}}(\theta, \eta, \zeta))\|^2 \\ &\qquad + \|K_\theta(\theta - \mu_\theta)\|^2 \end{aligned} \tag{7}$$

where μ_θ denotes the prior mean of the surface reflectance and SIF, while K_e and K_θ denote the Cholesky factors of the inverse of the noise covariance matrix and the inverse of the prior covariance matrix, respectively.

In this work, the prior model for the surface reflectance and SIF is generated by drawing random realizations from functions that model the general shape of the surface reflectance (a logistic function) and SIF (a sum of two Gaussians) spectra, as presented in [18]. The prior model construction involves computations of sample statistics of these samples which are then used to create a Gaussian prior model. In the prior model, we approximated the ρ_{surf} and F as mutually uncorrelated.

B. AE Method for SIFFI Retrieval

In the construction of the posterior model (5) and the MAP estimation (7), the auxiliary parameters (η, ζ) are treated as known fixed parameters and uncertainty related to the unknown and uninteresting measurement noise e is marginalized out from the model (see [24], [25] for details). However, in practice, the geometry η is well-known, but the atmospheric state ζ is most likely not accurately known, leading to model errors in the estimation of SIF. In such a situation, the ideal case would be to also marginalize the uncertainties related

to other inaccurately known auxiliary parameters, such as ζ, from the model, but this marginalization is a high-dimensional integration problem that does not usually have a closed form or computationally feasible solution. In the AE method, the idea is to treat the uncertainties related to the unknown auxiliary parameters by an approximate marginalization, where the model errors due to the uncertainty in the values of atmospheric state ζ are propagated into an additive AE noise process and the posterior is marginalized approximately over the AE noise (for detailed explanation see [24], [25]).

To explain the approach in the SIFFI setup, we write the observation model as

$$\begin{aligned} H_{\text{obs}} &= L_{\text{sim}}(\theta, \eta, \zeta) + e \\ &= L_{\text{sim}}(\theta, \eta, \zeta_0) + [L_{\text{sim}}(\theta, \eta, \zeta) - L_{\text{sim}}(\theta, \eta, \zeta_0)] + e \\ &= L_{\text{sim}}(\theta, \eta, \zeta_0) + \xi + e \end{aligned} \tag{8}$$

where ζ_0 is the fixed value that we use for the parameter vector ζ in the retrieval of θ, and ξ is the AE due to the possible deviation in the ζ_0 from the unknown true atmospheric state ζ.

Marginalizing the likelihood model with respect to ζ and e, we obtain [24], [25]

$$p(H_{\text{obs}}|\theta) = \int p_e(H_{\text{obs}} - L_{\text{sim}}(\theta, \eta, \zeta_0) - \xi) p_{\xi|\theta}(\xi|\theta) d\xi \tag{9}$$

when we assume that e and ξ are mutually independent. In general, the integral in (9) does not have a closed-form solution. However, by assuming Gaussian models for p_e and $p_{\xi|\theta}$, we obtain closed-form approximation for the integral. Let

$$p(\xi, \theta) \propto \exp\left(-\frac{1}{2}\begin{bmatrix}\xi - \mu_\xi \\ \theta - \mu_\theta\end{bmatrix}^{\text{T}} \begin{bmatrix}\Gamma_\xi & \Gamma_{\xi\theta} \\ \Gamma_{\theta\xi} & \Gamma_\theta\end{bmatrix}^{-1} \begin{bmatrix}\xi - \mu_\xi \\ \theta - \mu_\theta\end{bmatrix}\right) \tag{10}$$

be the Gaussian approximation of the joint density of ξ and θ, and $e \sim \mathcal{N}(\mu_e, \Gamma_e)$. Then, we get the conditional density

$$\xi|\theta \sim \mathcal{N}(\mu_{\xi|\theta}, \Gamma_{\xi|\theta}) \tag{11}$$

where

$$\begin{aligned} \mu_{\xi|\theta} &= \mu_\xi + \Gamma_{\xi\theta}\Gamma_\theta^{-1}(\theta - \mu_\theta) \quad \text{and} \\ \Gamma_{\xi|\theta} &= \Gamma_\xi - \Gamma_{\xi\theta}\Gamma_\theta^{-1}\Gamma_{\theta\xi}. \end{aligned} \tag{12}$$

Now we can define a new Gaussian random variable ν such that

$$\nu|\theta = e + \xi|\theta \quad \text{and} \quad \nu|\theta \sim \mathcal{N}(\mu_{\nu|\theta}, \Gamma_{\nu|\theta}) \tag{13}$$

where

$$\mu_{\nu|\theta} = \mu_e + \mu_{\xi|\theta} \quad \text{and} \quad \Gamma_{\nu|\theta} = \Gamma_e + \Gamma_{\xi|\theta}. \tag{14}$$

Using the Gaussian models for p_e and $p_{\xi|\theta}$, the integral in (9) becomes convolution integral of two Gaussian densities, leading to likelihood density

$$H_{\text{obs}}|\theta \sim \mathcal{N}(H_{\text{obs}} - L_{\text{sim}}(\theta, \eta, \zeta_0) - \mu_{\nu|\theta}, \Gamma_{\nu|\theta}) \tag{15}$$

and the MAP estimate leads to

$$\hat{\theta}_{\text{MAP}} = \arg\min_\theta \ \|K_{\nu|\theta}(H_{\text{obs}} - L_{\text{sim}}(\theta, \eta, \zeta_0) - \mu_{\nu|\theta})\|^2 + \|K_\theta(\theta - \mu_\theta)\|^2 \tag{16}$$

where $\mu_{\nu|\theta}$ denotes the AE sample mean, $K_{\nu|\theta}$ denotes the Cholesky factor of $\Gamma_{\nu|\theta}^{-1}$, and ζ_0 denotes the fixed realization for the auxiliary parameter. Remark that the MAP estimate with the AE is of similar form as the conventional MAP estimate (7) but with updated noise mean and Cholesky corresponding to the combined noise process $e + \xi$ in the model (8).

In practice, to estimate μ_ξ and Γ_ξ, we need a prior model for the primary unknowns ρ_{surf} and F and for the auxiliary unknown ζ. Then we can generate an ensemble of the AEs by drawing N random samples $\{\theta^{(k)}, \zeta^{(k)}\}$ from the prior models, computing the AE noise realizations

$$\xi^{(k)} = L_{\text{sim}}(\theta^{(k)}, \eta, \zeta^{(k)}) - L_{\text{sim}}(\theta^{(k)}, \eta, \zeta_0), \quad k = 1, \ldots, N \tag{17}$$

and then computing the sample mean and covariance from the ensemble $\{\xi^{(k)}\}$.

Another approximation we do is that we set the cross-covariance term $\Gamma_{\xi\theta}$ to zero since in most applications this approximation has been shown to not affect the estimation results substantially while simultaneously requiring a significantly smaller set of samples [19], [20], [24], [25]. This approximation will simplify the AE terms shown in (14) to

$$\mu_{\nu|\theta} \approx \mu_e + \mu_\xi \quad \text{and} \quad \Gamma_{\nu|\theta} \approx \Gamma_e + \Gamma_\xi. \tag{18}$$

C. Test Datasets

In this study, we consider a fixed observation and illumination geometry to demonstrate the AE approach since it is expected that the AE method works similarly regardless of the geometry. The observing instrument at TOA has a nadir view geometry and the solar zenith angle was set at 40°. To significantly speed up forward model evaluations, we use a high spectral resolution LUT containing radiative transfer simulations in the spectral range of 650–785 nm. In the forward model computations, the transfer functions in the LUT are spectrally convolved to a simulated high spectral resolution instrument with a Gaussian slit function with a full width at half maximum of 0.25 nm and an output spectral sampling interval of 0.33 nm. The LUT covers a range of values for AOD at 550 nm with a range of 0.03–0.60, TCWV with a range of 1.9–3.0 g/cm^2, aerosol scattering asymmetry parameter with a range of 0.55–0.80, SSA with a range of 0.8–1.0, and Ångström exponent with a range of 0.02–2.00. These five parameters are the elements of the parameter vector ζ that characterize the atmospheric state as they affect the forward model prediction of the TOA signal, in addition to the primary unknowns ρ_{surf} and F, at the fluorescence emission spectral range.

To simulate a realistic scenario H_{obs}, the simulated spectral radiance L_{sim} of (1) is corrupted by adding a normally distributed random error e

$$H_{\text{obs}} = L_{\text{sim}} + e \tag{19}$$

where $e \sim \mathcal{N}(0, a \cdot L_{\text{sim}} + b)$ with coefficients a and b as $9.009 \cdot 10^{-5}$ [mW/(m^2 · sr · nm)] and $3.660 \cdot 10^{-4}$ [mW/(m^2 ·

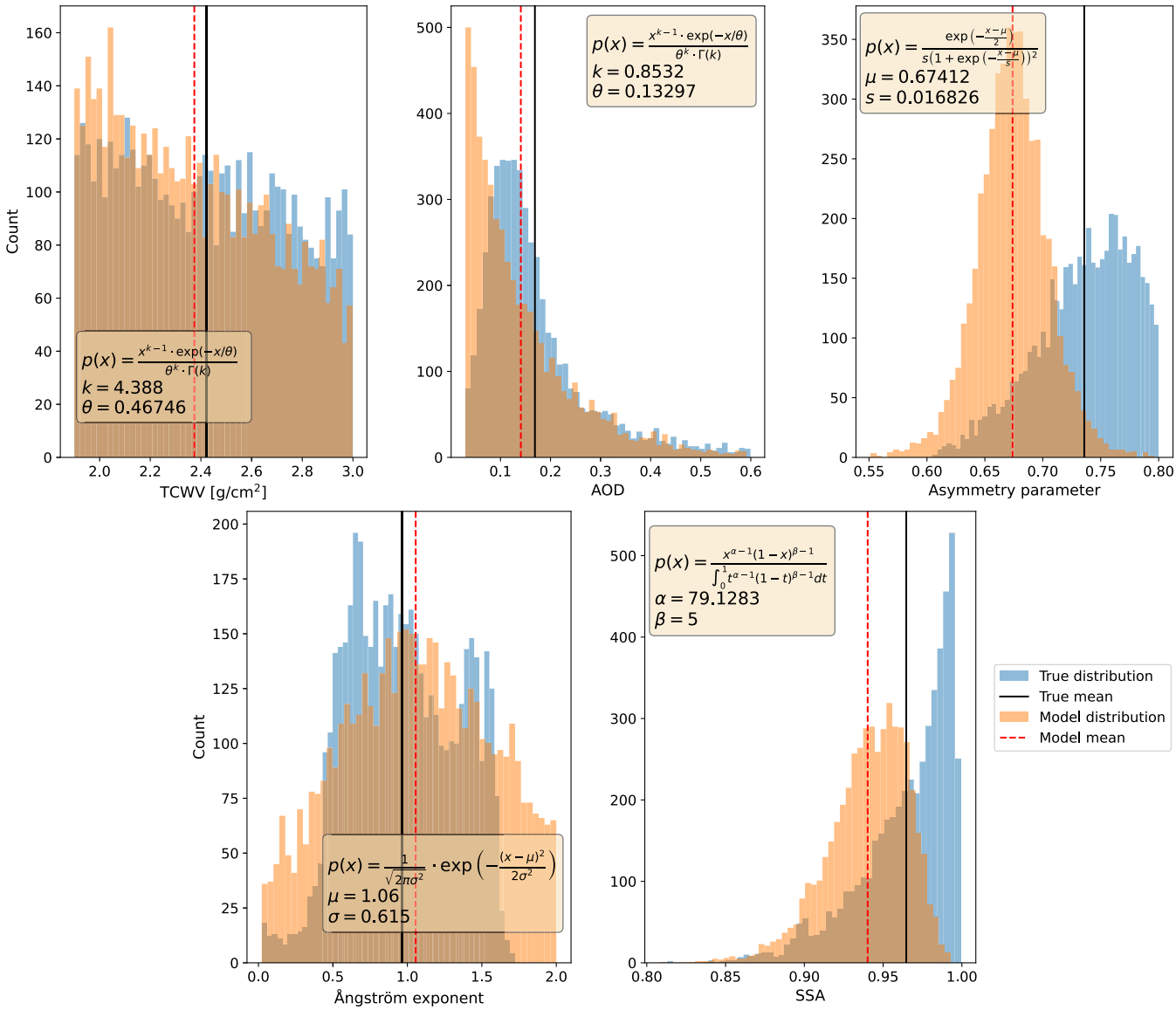


Fig. 1. Histograms with 5000 randomly drawn realizations from the prior model and the true distributions for unknown atmospheric parameters. The equations are for the prior models and they are gamma distribution for TCWV and AOD, logistic distribution for asymmetry parameter, Gaussian distribution for the Ångström exponent, and beta distribution for SSA.

sr·nm)]^2, respectively, as described for the FLORIS instrument in [26].

For the true target values of ρ_{surf} and F, we use leaf-level passive remote sensing measurement data of surface reflectance and SIF from the BIOHYPE dataset [27] as presented in [18]. The BIOHYPE dataset consists of four tree species (*Celtis australis*, *Morus alba*, *Platanus* × *acerifolia*, and *Phoenix canariensi*) commonly encountered in Spain in the city of Valencia. The dataset has a total of 291 paired ρ_{surf} and F spectra at 1-nm spectral sampling interval measured under natural illumination conditions. This dataset was chosen in this proof-of-concept simulation study because no direct measurements of full-fluorescence spectra from the TOA level are available. Likewise, for the TOC level, SIF measurements are either retrieved using indirect methods or model-simulated. As for the true values for the atmospheric state ζ, we use data from the global CAMS reanalysis dataset [28] using four different date indexes and two different times. The chosen dates were the 15th day of January, April, July, and October of the year 2023, and the times were midnight and noon (UTC).

The prior distribution models for each of the aerosol and water vapor atmospheric parameters are based on data from the aerosol climatology database MACv2 [29] and from [30], respectively. These distributions are shown in Fig. 1 along with the visualizations of the true probability distributions. To compute AE statistics, we draw 2000 random samples from the corresponding prior model for each of the atmospheric parameters. We use the mean of the prior model as the fixed realization ζ_0 in the AE model and MAP estimation (16).

To get a measure for the error between the true values and the fixed realization for the atmospheric parameters, we define a parameter distance metric

$$d(\zeta_{\text{true}}) = \sqrt{\sum_{i=1}^{5} \left(\frac{\zeta_{\text{true}}^{(i)} - \zeta_0^{(i)}}{\hat{\sigma}_\zeta^{(i)}} \right)^2} \tag{20}$$

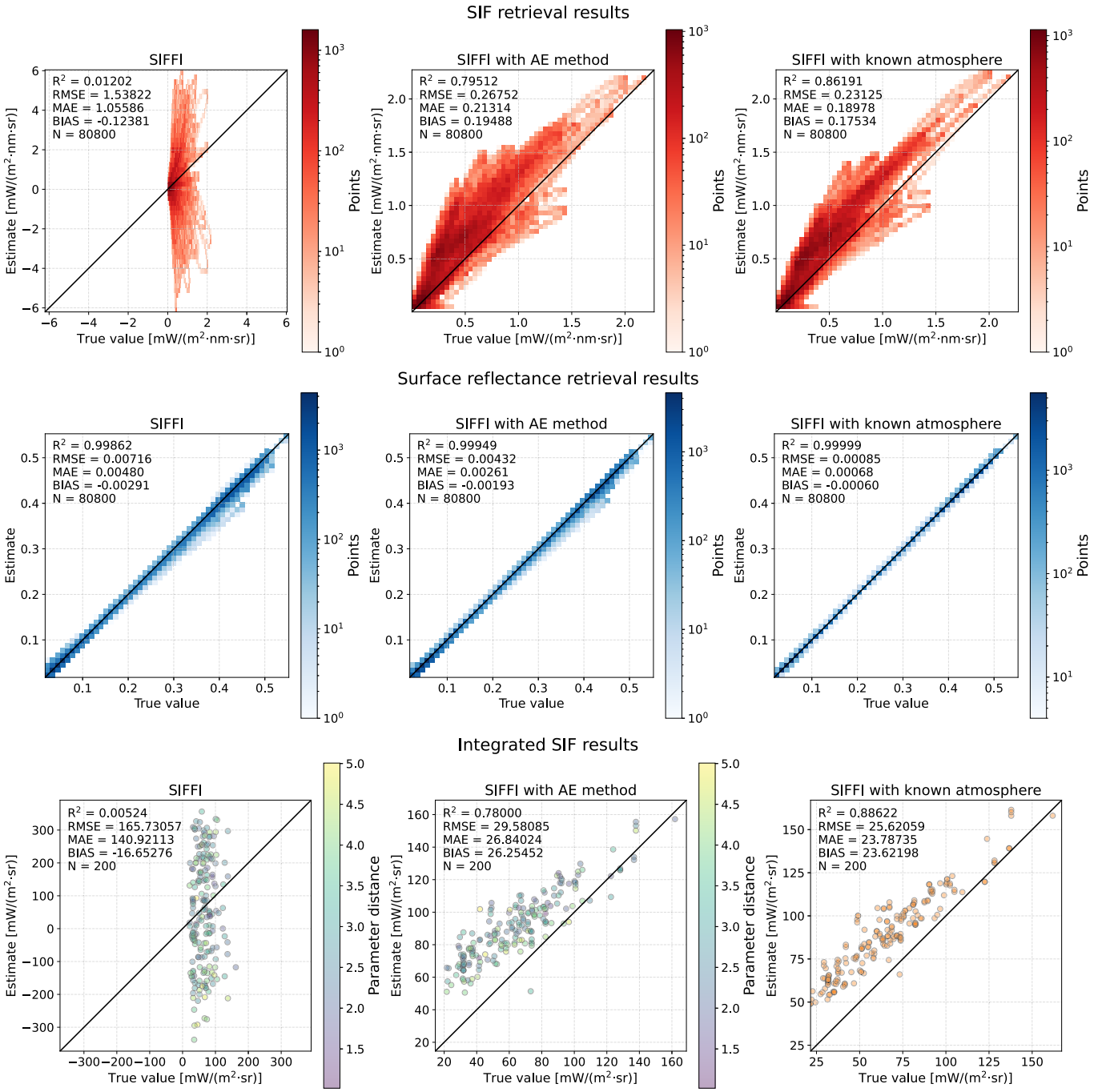


Fig. 2. (Top) Comparison between the SIF estimates per wavelength or instrument channel and (middle) surface reflectance against the true values. (Bottom) Comparison between the estimated and true SIF values integrated over the wavelength range of 650–785 nm. The left column panels show the results using SIFFI with incorrect atmospheric parameters, while the middle column panels show the results when applying the AE method in the SIFFI retrieval. The right column panels show the results using SIFFI in the ideal situation of a known atmosphere. The black line in the panels is the 1:1 line.

where $\zeta_{\text{true}}^{(i)}$ is the true value for the atmospheric parameter i and $\hat{\sigma}_{\zeta}^{(i)}$ is the standard deviation computed from the ensemble of the true values for the atmospheric parameter i.

III. Results and Discussion

Fig. 2 shows density scatter plots of true and estimated SIF (top row) and surface reflectance (middle row) for all wavelengths (650–785 nm) from the test set of 200 SIF and surface reflectance spectral samples when all five atmospheric parameters are unknown and fixed to their prior means. This corresponds to a scenario where no auxiliary information sources are available to provide information about the atmospheric state. In addition to density scatter plots, Fig. 2 (third row) shows scatter plots of the estimated SIF values against the true values when the SIF values for every individual retrieved spectra are integrated over the wavelength range of 650–785 nm using the trapezoidal rule. The purpose of integrating over the spectral range of interest is to get a measure of the total SIF emission error. The parameter distance metric in the figure is computed with (20). The figure shows that using the AE

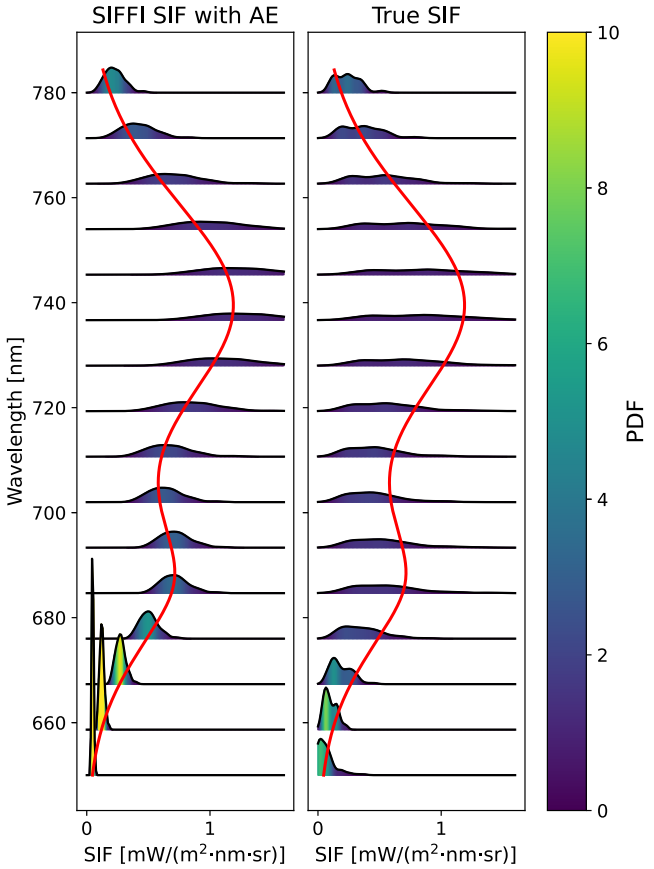


Fig. 3. Probability density of the estimated SIF with (left) AE method and (right) true values at different wavelengths where the red curve shows the prior mean of SIF across 650–785 nm.

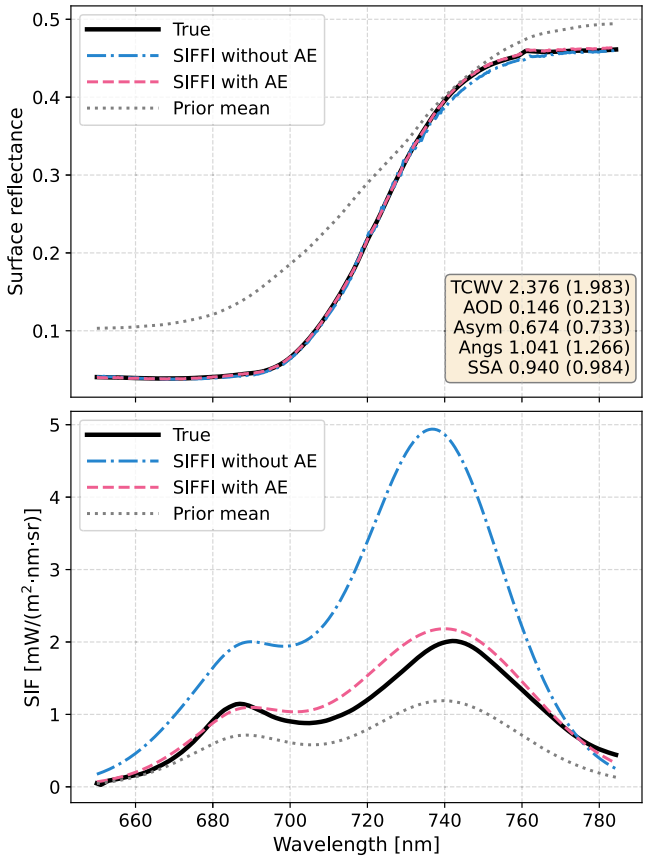


Fig. 4. Example of full spectrum retrieval of (top) surface reflectance and (bottom) SIF with and without the AE method. The true values for the atmospheric parameters are inside parenthesis.

method, the estimation accuracy increases significantly when comparing the results against the regular SIFFI estimates with incorrect atmosphere-related information. The panels on the right column with the fixed but correct atmospheric state are added here to show the SIFFI estimates as a reference of how estimation using correct atmospheric parameters performs. From the integral plot, we can see that there is no clear connection between the error in the estimation with regular SIFFI and parameter distance from the truth. This indicates that the error in some of the unknown atmospheric parameters affects the retrieval performance more than others, which is as expected.

To validate the justification for the approximation done in (18), we tested running the computations without neglecting the cross-covariance term $\Gamma_{\xi\theta}$. As expected, the results (shown in Fig. S1 in the Supplements) had slightly better accuracy metrics when compared against the results without the cross-covariance term shown in Fig. 2. However, using the full AE model required increasing the number of AE noise realizations from 2000 to 50 000 samples, since using fever samples caused numerical issues due to poorly converged cross-covariance estimates.

Since the information about the atmospheric state comes from a prior model, there exists a reasonable concern that SIFFI always returns just the prior mean of the primary unknowns. However, from Fig. 3, we can see that this is not the case since there is no clear singular peak in the probability density function around the prior mean. Fig. 4 shows an example of a retrieved surface reflectance and SIF spectra with and without the AE method. The figure shows quite clearly how the uncertainty in the atmospheric state affects the SIF retrieval accuracy without the AE method. The corresponding transfer functions (L_0, E_{toc}, $T^{\uparrow}$, and S) for the case shown in Fig. 4 are shown in Fig. S2 in the Supplements.

In addition to testing when all five atmospheric parameters are unknown, we also investigated cases where only one atmospheric parameter is unknown while the others are known since, quite often, it is possible to get prior information about the atmospheric parameters before a retrieval using data, for example, from other satellite retrievals, forecasting, and climatology. The parameters we tested to be unknown were AOD and TCWV since they are the most driving parameters for aerosol and gas presence, respectively, across the red and near-infrared region [16].

Fig. 5 shows density scatter plots of true and estimated SIF for all individual spectral channels when the AOD (left column) and the TCWV (right column) are unknown while all other atmospheric parameters are known. The figure shows quite clearly that without knowledge about either of the two parameters, the regular SIFFI is not able to accurately retrieve SIF. The error arising from incorrect assumptions about the atmospheric state can be seen from both the parameters, but the error is more evident when the TCWV is incorrect, affecting the red-edge region [16]. The figure also shows that using

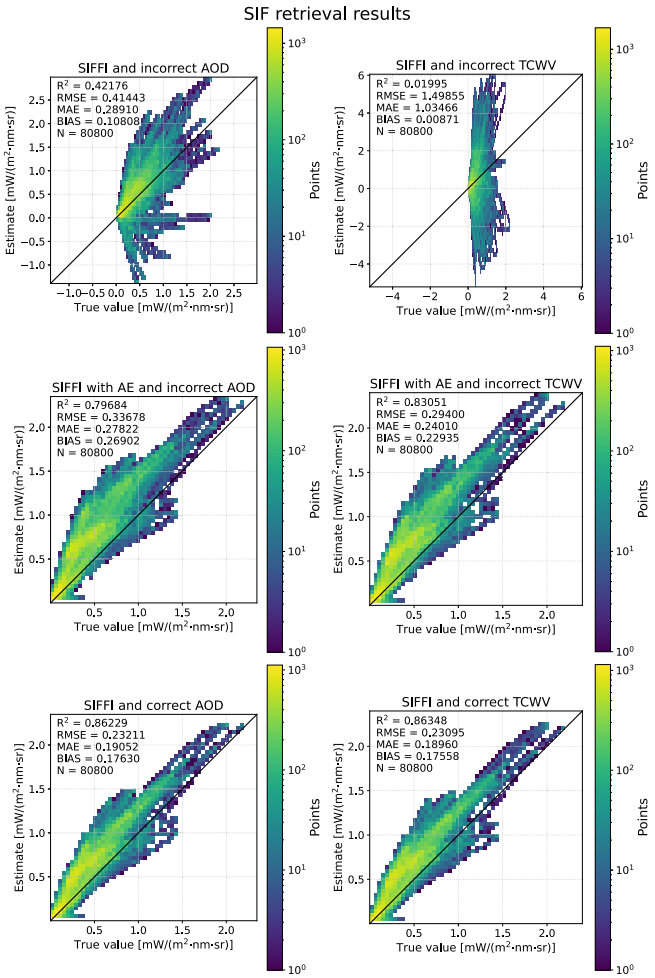


Fig. 5. Comparison between SIF estimates per wavelength or instrument channel against the true values when (left) the AOD and (right) the TCWV are the poorly characterized atmospheric parameters. The top row panels show the results using SIFFI with one incorrect atmospheric parameters. The middle row panels show the results when applying the AE method in the SIFFI retrieval. The bottom row panels show the results using SIFFI in the ideal situation of a known atmosphere. The black line in the panels is the 1:1 line.

the AE method with SIFFI, the estimation accuracy improves significantly.

Deeper analysis on the parameter-specific sensitivities to different spectral regions could be beneficial to identify the variables that have the most significant contributions to SIF retrieval accuracy (check Figs. S3 and S4 in the Supplements). This information could then be used to study the error sensitivity of different parameters in the assumed atmospheric state in different spectral bands. Among the spectral regions relevant for SIF retrieval, the red and far-red peaks are of particular interest. The retrieval results at the approximate wavelength location of red (687 nm) and far-red (740 nm) SIF peaks are shown in Figs. S5–S7 in the Supplements. The results display similar patterns as observed in Figs. 2 and 5, reinforcing the consistency of the SIFFI retrieval algorithm and the AE method.

For satellite data applications, using the AE method can potentially enhance retrieval accuracy in the presence of nonneglectable uncertainties in the observed atmospheric state. In our study, the measurement geometry η is not marginalized out in (16). Thus, the AE terms (mean and covariance) are dependent on the geometry. For real-world applications, the AE terms could be precomputed with a range of different illumination and observation geometry values, therefore expanding their applicability. In such a scenario, it should be possible to then use the AE terms, e.g., stored in an LUT, which closely match the actual measurement geometry. This storing of values should be computationally feasible for LUT-based retrieval algorithms since the same LUT that is used in the retrieval can be used to compute the required samples for the AE method. Once the samples are generated, computing the AE terms of (18) is a trivial task in terms of computing power. However, for methods that use, for example, full radiative transfer simulations, computing the necessary number of samples for the successful use of the AE method may have high computational costs. Conveniently, the computations are required to be carried only once in the precomputation step.

The simulation test cases were designed to demonstrate the effectiveness of the AE approach in marginalizing uncertainties from atmospheric characterization. Atmospheric variability was realistically captured using distributions from the CAMS service, while a low-accuracy atmospheric characterization (based on mean values of appropriate distributions) was assumed to create a challenging scenario. Instrument effects were simplified by assuming zero-bias, spectrally uncorrelated Gaussian sensor noise, and a spectral resolution representative of current and future SIF-dedicated hyperspectral sensors (e.g., GOSAT, OCO-2, FLEX). Although this setup may be optimistic in terms of sensor noise, it isolates the influence of atmospheric uncertainty. Thus, while the final accuracy on real-satellite applications will depend on different instrument specifications or canopy-related characteristics, the here proved approach's capability to address atmospheric uncertainties remains broadly applicable.

IV. Conclusion

In this study, we augmented the SIFFI retrieval with the AE method to marginalize approximately over uncertainty in the atmospheric state. The AE method, grounded in a Bayesian framework for inverse problems, has proven effective in reducing estimation errors arising from modeling inaccuracies and uncertainties across various applications. Here, we demonstrate the AE method's potential to enhance total SIF retrieval estimates at the satellite level when coupled with the SIFFI technique, mitigating errors introduced by unknown or poorly estimated atmospheric parameters. The enhanced quality and consistency of SIF estimates can provide a more reliable proxy for photosynthetic activity, supporting carbon cycle research and improving the interpretation of vegetation dynamics at global scales.

Future work will focus on validating the SIFFI retrieval algorithm using data from existing instruments with capability to retrieve SIF and data from the upcoming ESA FLEX mission. We also plan to further test the limits of the AE method, since the AE method has the potential for integration to other SIF retrieval approaches, providing a robust pathway to mitigate atmospheric disturbances and enhance SIF retrieval accuracy across diverse atmospheric observational conditions.

APPENDIX

Nomenclature contains a summary of the key acronyms used in this article.

ACKNOWLEDGMENT

The authors would like to thank Shari Van Wittenberghe and the BIOHYPE Project for kindly providing the data for the test set in this article. They declare no competing interests.

REFERENCES

[1] M. Meroni et al., "Remote sensing of solar-induced chlorophyll fluorescence: Review of methods and applications," *Remote Sens. Environ.*, vol. 113, no. 10, pp. 2037–2051, Oct. 2009. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S003442570900162X

[2] L. Guanter et al., "Global and time-resolved monitoring of crop photosynthesis with chlorophyll fluorescence," *Proc. Nat. Acad. Sci. USA*, vol. 111, no. 14, pp. E1327–E1333, Apr. 2014, doi: 10.1073/pnas.1320008111.

[3] G. H. Mohammed et al., "Remote sensing of solar-induced chlorophyll fluorescence (SIF) in vegetation: 50 years of progress," *Remote Sens. Environ.*, vol. 231, Sep. 2019, Art. no. 111177. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0034425719301816

[4] J. Joiner, Y. Yoshida, A. P. Vasilkov, Y. Yoshida, L. A. Corp, and E. M. Middleton, "First observations of global and seasonal terrestrial chlorophyll fluorescence from space," *Biogeosciences*, vol. 8, no. 3, pp. 637–651, Mar. 2011. [Online]. Available: https://bg.copernicus.org/articles/8/637/2011/

[5] C. Frankenberg et al., "New global observations of the terrestrial carbon cycle from GOSAT: Patterns of plant fluorescence with gross primary productivity," *Geophys. Res. Lett.*, vol. 38, no. 17, Sep. 2011, Art. no. L17706, doi: 10.1029/2011GL048738.

[6] J. A. Plascyk, "The MK II Fraunhofer line discriminator (FLD-II) for airborne and orbital remote sensing of solar-stimulated luminescence," *Opt. Eng.*, vol. 14, no. 4, Aug. 1975, Art. no. 144339, doi: 10.1117/12.7971842.

[7] J. A. Plascyk and F. C. Gabriel, "The Fraunhofer line discriminator MKII—An airborne instrument for precise and standardized ecological luminescence measurement," *IEEE Trans. Instrum. Meas.*, vol. IM-24, no. 4, pp. 306–313, Nov. 1975.

[8] M. Gunson and A. Eldering, "OCO-2 level 2 bias-corrected solar-induced fluorescence and other select fields from the IMAP-DOAS algorithm aggregated as daily files, Retrospective processing V10r, Greenbelt, MD, USA, Goddard Earth Sciences Data and Information Services Center (GES DISC)," Data Inf. Serv. Cent. GES DISC [Data Set], Greenbelt, MD, USA, Tech. Rep. 10r, 2020, doi: 10.5067/XO2LBBNPO010.

[9] M. Gunson and A. Eldering, "OCO-3 level 2 bias-corrected solar-induced fluorescence and other select fields from the IMAP-DOAS algorithm aggregated as daily files, Retrospective processing V10r, Greenbelt, MD, USA, Goddard Earth Sciences Data and Information Services Center (GES DISC)," Data Inf. Serv. Cent. GES DISC, Greenbelt, MD, USA, Tech. Rep. 10r, 2021, doi: 10.5067/NOD1DPPBCXSO.

[10] C. Frankenberg, "GOSAT solar-induced chlorophyll fluorescence datasets 2009–2020," California Inst. Technol. (Caltech), Pasadena, CA, USA, Tech. Rep. 1, Jan. 2022.

[11] P. Köhler, M. J. Behrenfeld, J. Landgraf, J. Joiner, T. S. Magney, and C. Frankenberg, "Global retrievals of solar-induced chlorophyll fluorescence at red wavelengths with TROPOMI," *Geophys. Res. Lett.*, vol. 47, no. 15, p. 2020, Aug. 2020, doi: 10.1029/2020GL087541.

[12] L. Guanter et al., "The TROPOSIF global sun-induced fluorescence dataset from the sentinel-5P TROPOMI mission," *Earth Syst. Sci. Data*, vol. 13, no. 11, pp. 5423–5440, Nov. 2021. [Online]. Available: https://essd.copernicus.org/articles/13/5423/2021/

[13] C. Frankenberg, A. Butz, and G. Toon, "Disentangling chlorophyll fluorescence from atmospheric scattering effects in O_2 A-band spectra of reflected sun-light," *Geophys. Res. Lett.*, vol. 38, no. 3, 2011, Art. no. L03801.

[14] L. Guanter et al., "Retrieval and global assessment of terrestrial chlorophyll fluorescence from GOSAT space measurements," *Remote Sens. Environ.*, vol. 121, pp. 236–251, Jun. 2012.

[15] F. Zhao, W. Ma, J. Zhao, Y. Guo, M. Tariq, and J. Li, "Global retrieval of the spectrum of terrestrial chlorophyll fluorescence: First results with TROPOMI," *Remote Sens. Environ.*, vol. 300, Jan. 2024, Art. no. 113903. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0034425723004546

[16] N. Sabater, P. Kolmonen, S. Van Wittenberghe, A. Arola, and J. Moreno, "Challenges in the atmospheric characterization for the retrieval of spectrally resolved fluorescence and PRI region dynamics from space," *Remote Sens. Environ.*, vol. 254, Mar. 2021, Art. no. 112226.

[17] M. Drusch et al., "The fluorescence explorer mission conceptESA's Earth explorer 8," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1273–1284, Dec. 2016.

[18] A. Kukkurainen, A. Lipponen, V. Kolehmainen, A. Arola, S. Cogliati, and N. Sabater, "SIFFI: Bayesian solar-induced fluorescence retrieval algorithm for remote sensing of vegetation," *Remote Sens. Environ.*, vol. 318, Mar. 2025, Art. no. 114558. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0034425724005844

[19] J. Kaipio and E. Somersalo, *Statistical and Computational Inverse Problems*. Cham, Switzerland: Springer, 2005.

[20] J. Kaipio and E. Somersalo, "Statistical inverse problems: Discretization, model reduction and inverse crimes," *J. Comput. Appl. Math.*, vol. 198, no. 2, pp. 493–504, Jan. 2007. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0377042705007296

[21] B. Sierk et al., "The Copernicus CO_2M mission for monitoring anthropogenic carbon dioxide emissions from space," in *Proc. Int. Conf. Space Opt. (ICSO)*, Jun. 2021, p. 128.

[22] D. Tanre, M. Herman, P. Y. Deschamps, and A. de Leffe, "Atmospheric modeling for space measurements of ground reflectances, including bidirectional properties," *Appl. Opt.*, vol. 18, no. 21, pp. 3587–3594, Nov. 1979. [Online]. Available: https://opg.optica.org/ao/abstract.cfm?URI=ao-18-21-3587

[23] L. Guanter et al., "Developments for vegetation fluorescence retrieval from spaceborne high-resolution spectrometry in the O2-A and O2-B absorption bands," *J. Geophys. Res., Atmos.*, vol. 115, no. D19, Oct. 2010, Art. no. D19303, doi: 10.1029/2009JD013716.

[24] V. Kolehmainen, T. Tarvainen, S. R. Arridge, and J. P. Kaipio, "Marginalization of uninteresting distributed parameters in inverse problems-application to diffuse optical tomography," *Int. J. Uncertainty Quantification*, vol. 1, no. 1, pp. 1–17, 2011.

[25] J. Kaipio and V. Kolehmainen, "Approximate marginalization over modelling errors and uncertainties in inverse problems," in *Bayesian Theory and Applications*. London, U.K.: Oxford Univ. Press, Jan. 2013, doi: 10.1093/acprof:oso/9780199695607.003.0032.

[26] W. Verhoef, C. van der Tol, and E. M. Middleton, "Hyperspectral radiative transfer modeling to explore the combined retrieval of biophysical parameters and canopy fluorescence from FLEX–Sentinel-3 tandem mission multi-sensor data," *Remote Sens. Environ.*, vol. 204, pp. 942–963, Jan. 2018. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0034425717303607

[27] S. Van Wittenberghe et al., "Upward and downward solar-induced chlorophyll fluorescence yield indices of four tree species as indicators of traffic pollution in Valencia," *Environ. Pollut.*, vol. 173, pp. 29–37, Feb. 2013.

[28] A. Inness et al., "The CAMS reanalysis of atmospheric composition," *Atmos. Chem. Phys.*, vol. 19, no. 6, pp. 3515–3556, 2019. [Online]. Available: https://acp.copernicus.org/articles/19/3515/2019/

[29] S. Kinne, "The MACv2 aerosol climatology," *Tellus B, Chem. Phys. Meteorol.*, vol. 71, no. 1, pp. 1–21, Jan. 2019.

[30] G. Anderson, S. Clough, F. Kneizys, J. Chetwynd, and E. Shettle, "AFGL atmospheric constituent profiles (0.120km)," Hanscom Air Force Base, Air Force Geophysics Laboratory, Bedford, MA, USA, Tech. Rep. AFGL-TR-86-0110, May 1986, p. 46.



Antti Kukkurainen received the M.Sc. degree in computational physics and the Ph.D. degree in applied physics from the University of Eastern Finland, Kuopio, Finland, in 2019 and 2025, respectively.

He is currently working as a Research Scientist at the Finnish Meteorological Institute, Kuopio. He is also the Principal Developer of the SIFFI retrieval algorithm. His research interests primarily focus on atmospheric radiative transfer models and remote sensing of solar-induced fluorescence.



Antti Lipponen received the M.Sc. degree in medical physics from the University of Kuopio, Kuopio, Finland, in 2008, and the Ph.D. degree in applied physics from the University of Eastern Finland, Kuopio, in 2014.

He is currently working as a Research Scientist at the Finnish Meteorological Institute, Kuopio. His research interests include atmospheric remote sensing, Bayesian inverse problems, and data analysis and modeling related to atmospheric and climate applications.



Neus Sabater received the Licentiate degree in physics, the M.Sc. degree in remote sensing, and the Ph.D. degree in remote sensing from the Universitat de València, Valencia, Spain, in 2010, 2012, and 2018, respectively.

Since 2012, she has been contributing to the European Space Agency (ESA) Fluorescence EXplorer (FLEX) mission, initially at the Universitat de València and, since 2017, at the Finnish Meteorological Institute (FMI), Finland. She is currently working as a Research Scientist at FMI, focusing on atmospheric correction, atmospheric radiative transfer, hyperspectral data processing, fluorescence retrieval, and arctic vegetation.

Dr. Sabater is the Principal Investigator of the ArcticSIF Project (funded by the Research Council of Finland) and plays a key role in developing the atmospheric correction algorithm of the FLEX ground processor under the ESA DISC initiative.



Antti Arola received the M.Sc.Tech. degree in hydrology from Helsinki University of Technology, Espoo, Finland, in 1991, the M.Sc. degree in civil engineering from the University of Washington, Seattle, WA, USA, in 1993, the Phil.Lic. degree in meteorology from the University of Helsinki, Helsinki, Finland, in 1999, and the Ph.D. degree in applied physics from the University of Kuopio, Kuopio, Finland, in 2006.

He is currently a Research Professor with the Finnish Meteorological Institute (FMI), Kuopio, and is leading the Atmospheric Radiation Group is a part of the FMI's Atmospheric Research Centre of Eastern Finland. His research interests include aerosol and cloud optical properties, their effects on radiative transfer and Earth's radiative balance, and aerosol–cloud interactions.



Ville Kolehmainen received the Ph.D. degree in applied physics from the University of Kuopio, Kuopio, Finland, in 2001.

He is a Professor of computational physics with the University of Eastern Finland, Kuopio. His research focuses on mathematical models and computational methods for inverse imaging problems with applications ranging from medical imaging to remote sensing.